

There Is No Such Thing as Science. There Are Only Scientists.

Ma ZhenYao

Tao recently raised a question I very much like while discussing the *misalignment* between AI and mathematics: when AI can produce answers with ever greater ease, why do we still do mathematics? In his *Mathematics in the Age of AI*, the goals he sets out for mathematics go far beyond solving open problems. They include developing new theories and structures, understanding the world, building a shared body of mathematical knowledge, and creating works of lasting value. One idea I take from this is that the answer itself is not mathematics; the answer is the reward for someone who has truly come to understand the landscape.

I want to go one step further: science itself is the reward for those who come to understand the oracular landscape.

If we imagine the experimenter as a mountaineer, then an experiment is an exploration of a mountain landscape. An adventurer first looks into the distance and chooses a destination: is there a mountain over there? What lies beyond it? If I follow this river upstream, will it lead to a place no one has ever reached before? Then he shoulders his pack and actually enters the forest, leaving coordinates along the way, checking his route, getting lost, turning back, and searching again, before finally returning to share with us the landscape he has seen.

Science is much the same. A scientist first asks a question, then enters it by rigorous, testable, and falsifiable means. Experiments, proofs, statistics, and replication are the coordinates he leaves behind. They matter not because the coordinates are themselves the landscape, but because only through them do we have reason to believe that this person truly went there - that what he brought back was not a hallucination, not an imagined map. The importance of science therefore lies both in which destination the scientist chose and in the rigor of the means by which he brought a previously invisible landscape before us.

This is also why, when we talk about how AI may change science, I often feel that we should first look back at art. Art has already lived through technological shocks of this kind again and again. After photography emerged in the nineteenth century, one of the functions painting had long performed - accurately recording faces, cities, wars, and reality - was, for the first time, taken over by machines on a large scale. Film, photography, and mechanical reproduction then continued to expand the boundaries of image production. If the value of painting were simply to "draw the world before our eyes accurately," the camera would of course be an almost brutal substitute.

Yet what endured was not a competition over "who could paint more accurately." The Impressionists began to pursue light, air, the colors of a fleeting moment, and the act of seeing itself. Monet could paint the same Rouen Cathedral again and again at different times, because what mattered was no longer merely "what does the cathedral look like?" but what, exactly, a person saw on a particular afternoon. Later came Van Gogh, Cezanne, and countless other artists who did not stop working simply because photography could reproduce the world more accurately. We remember not only the images they ultimately produced, but also how they saw the world, how they chose, how they struggled, and the path they spent a lifetime walking.

Technology, then, did not simply "destroy art." What it often destroyed was only a scarce ability that had previously been mistaken for art itself. Once accurate reproduction of reality ceased to be scarce, we could see more clearly that art had never been merely the reproduction of reality. Something similar may now be happening in science. When proofs, code, papers, and even the answer to a question begin to become cheap, we are forced to ask again: what did we truly value all along - these outputs, or the understanding, judgment, questions, and exploration that gave rise to them?

At the beginning of *The Story of Art*, Gombrich wrote what later became a famous sentence: "There really is no such thing as Art. There are only artists." He later explained that the point was to bring the abstract, capital-A "Art" back down to concrete human beings: art is not some mysterious entity detached from people. What truly exists are individual artists facing specific problems, mastering specific skills, and making specific choices.

I increasingly feel that science may be the same.

Today we are accustomed to imagining science as a vast machine of knowledge: papers, theorems, data, benchmarks, and experimental results are continuously added to it, and Science itself seems to move forward. But the person who asks a question truly worth pursuing, the person who decides which mountain to walk toward, the person who sees a new landscape for the first time, has always been a particular human being. A paper is only the map he leaves behind after returning; an answer is only the reward received upon reaching a summit.

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